

SmartDoseAI Clinical Report

1. Patient Details

Patient Name: neha kulkarni
Age: 47 | Gender: Male | Blood Group: A+
Assigned Doctor: Dr. Priya
Appointment Date: 2026-05-09

2. Original Lab Report Information

Neha_Kulkarni.pdf (Uploaded by Reception)

3. Confirmed Laboratory Values

- Fasting Glucose: 196 mg/dL
- HbA1c: 9.5%
- Insulin: 100 µU/mL
- BMI: 30.2 kg/m²
- Creatinine: Not recorded
- BUN: Not recorded
- eGFR: Not recorded
- Potassium: Not recorded
- Hemoglobin: Not recorded
- Blood Pressure: Not recorded
- Cholesterol: Not recorded
- Heart Rate: Not recorded

4. AI Prediction

Prediction: Type 2 Diabetes
Model Probability: N/A%
Confidence Band: High

5. Explainability (Model Insights)

Patient Explanation:
Your insulin, bmi, hba1c, post-meal glucose, and fasting glucose measurements were among the factors influencing the AI prediction.

Some measurements were not recorded, so the explanation is based only on the available information.

Main Contributing Factors:

- Insulin (100 µU/mL) -- Influence: Strong
- BMI (30.2 kg/m²) -- Influence: Moderate
- HbA1c (9.5%) -- Influence: Mild
- Post-meal Glucose (314 mg/dL) -- Influence: Mild
- Fasting Glucose (196 mg/dL) -- Influence: Mild
- Age (47 years) -- Influence: Mild
- Gender (Male) -- Influence: Mild

6. Smart Dose Recommendation

Recommended Regimen: Metformin 1000mg twice daily

Dose: 1000mg BID (Max Ceiling: 2550mg/day)

Organ Safety Assessment: Moderate Risk

Kidney Check: ? Unable to fully assess kidney safety. Required kidney-function data were not available. D

Heart Check: Standard monitoring

Interactions: None highlighted

Clinician Review Status: Requires Doctor Review

7. What-If Digital Twin Simulation

WARNING: What-if simulation - not a forecast of the patient's actual medical outcome.

Parameter Shifts:

- Fasting Glucose: 196 mg/dL -> 176 mg/dL (Simulated)

Actual Prediction: Type 2 Diabetes | Simulated: Type 2 Diabetes

Actual Dose: Metformin 1000mg twice daily (1000mg BID) | Simulated: Metformin 1000mg twice daily (1000mg BID)

Actual Kidney: Glucose-related kidney risk indicator: Elevated Note: This is a model/rule-based simulation, not a forecast of actual outcomes.

Actual Heart: Looks steady in this simulation | Simulated: Looks steady in this simulation

8. Clinical Guidance & Disclaimer

This report is generated by SmartDoseAI as an educational decision-support demo. It does not constitute autonomous medical advice or prescribing authority. All recommendations must be evaluated and approved by a licensed clinician prior to care implementation.